

A Standardized Benchmark for Skeleton-Based Rehabilitation Assessment Using Deep Learning

Ali Ismail-Fawaz¹, **Maxime Devanne**¹, Stefano Berretti², Jonathan Weber¹,
Germain Forestier¹

¹Université Haute-Alsace, IRIMAS

²Università di Firenze, MICC



The 20th IEEE International Conference on Automatic Face and Gesture Recognition

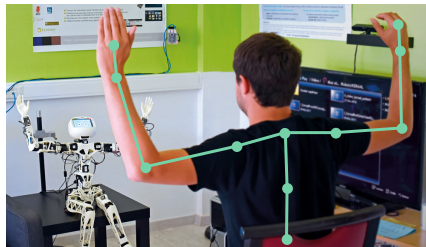
May 27th, 2026, Kyoto, Japan 門

Context and Motivation

- ▷ Rehabilitation assessment is critical for :
 - ▷ Monitoring patient recovery
 - ▷ Evaluating exercise quality
 - ▷ Supporting personalized therapy
- ▷ Video-based skeleton extraction enables :
 - ▷ Low-cost assessment
 - ▷ Remote rehabilitation
 - ▷ Smartphone/webcam deployment
- ▷ Deep learning methods are increasingly used for skeleton-based rehabilitation assessment

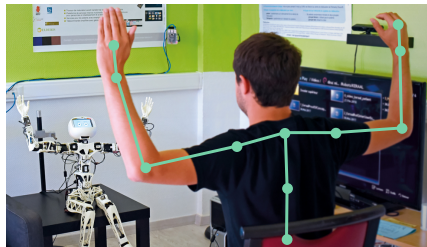
Context and Motivation

- ▷ Rehabilitation assessment is critical for :
 - ▷ Monitoring patient recovery
 - ▷ Evaluating exercise quality
 - ▷ Supporting personalized therapy
- ▷ Video-based skeleton extraction enables :
 - ▷ Low-cost assessment
 - ▷ Remote rehabilitation
 - ▷ Smartphone/webcam deployment
- ▷ Deep learning methods are increasingly used for skeleton-based rehabilitation assessment



Context and Motivation

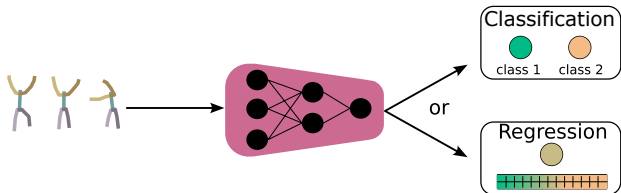
- ▷ Rehabilitation assessment is critical for :
 - ▷ Monitoring patient recovery
 - ▷ Evaluating exercise quality
 - ▷ Supporting personalized therapy
- ▷ Video-based skeleton extraction enables :
 - ▷ Low-cost assessment
 - ▷ Remote rehabilitation
 - ▷ Smartphone/webcam deployment
- ▷ Deep learning methods are increasingly used for skeleton-based rehabilitation assessment



Problem Statement

Rehabilitation Assessment

- ▷ Evaluates movement quality
- ▷ Classification example :
 - ▷ Correct vs incorrect squat
- ▷ Extrinsic regression example :
 - ▷ Score of correctness



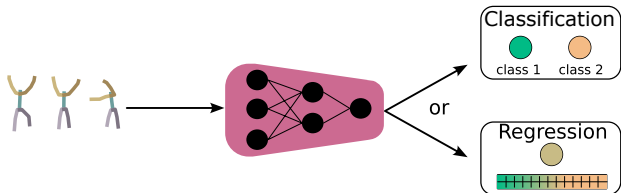
Main Issues in the Literature

- ▷ Evaluations are performed on very few datasets
 - ▷ Mainly KIMORE and UI-PRMD
- ▷ Inconsistent experimental protocols
- ▷ Limited reproducibility and unavailable code

Problem Statement

Rehabilitation Assessment

- ▷ Evaluates movement quality
- ▷ Classification example :
 - ▷ Correct vs incorrect squat
- ▷ Extrinsic regression example :
 - ▷ Score of correctness



Main Issues in the Literature

- ▷ Evaluations are performed on very few datasets
 - ▷ Mainly KIMORE and UI-PRMD
- ▷ Inconsistent experimental protocols
- ▷ Limited reproducibility and unavailable code

Contributions

▷ **Rehab-Pile :**

- ▷ Unified archive of skeleton-based rehabilitation datasets
- ▷ 60 datasets aggregated from 8 repositories
- ▷ Classification + extrinsic regression tasks

▷ **Unified Benchmarking Framework :**

- ▷ Fair comparison of 7 deep learning architectures
- ▷ Standardized preprocessing and evaluation
- ▷ Performance and efficiency analysis

▷ **Reproducibility :**

- ▷ Public release of datasets and code

Contributions

▷ **Rehab-Pile :**

- ▷ Unified archive of skeleton-based rehabilitation datasets
- ▷ 60 datasets aggregated from 8 repositories
- ▷ Classification + extrinsic regression tasks

▷ **Unified Benchmarking Framework :**

- ▷ Fair comparison of 7 deep learning architectures
- ▷ Standardized preprocessing and evaluation
- ▷ Performance and efficiency analysis

▷ **Reproducibility :**

- ▷ Public release of datasets and code

Contributions

▷ **Rehab-Pile :**

- ▷ Unified archive of skeleton-based rehabilitation datasets
- ▷ 60 datasets aggregated from 8 repositories
- ▷ Classification + extrinsic regression tasks

▷ **Unified Benchmarking Framework :**

- ▷ Fair comparison of 7 deep learning architectures
- ▷ Standardized preprocessing and evaluation
- ▷ Performance and efficiency analysis

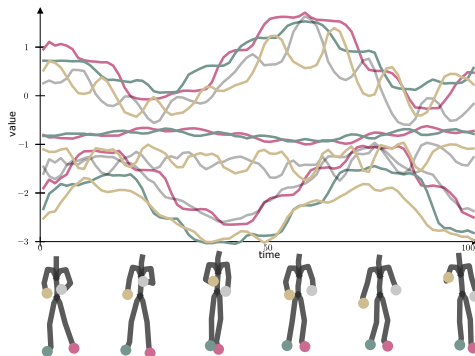
▷ **Reproducibility :**

- ▷ Public release of datasets and code

Skeleton Sequence Representation

- ▷ Input : sequence of human skeletons extracted from video
- ▷ Each skeleton contains J joints in D dimensions (2D, 3D or more)
- ▷ A sequence is represented as a multivariate time series of length L :

$$S \in \mathbb{R}^{L \times J \times D}$$



The Rehab-Pile

- ▷ We aggregated 8 existing repositories :

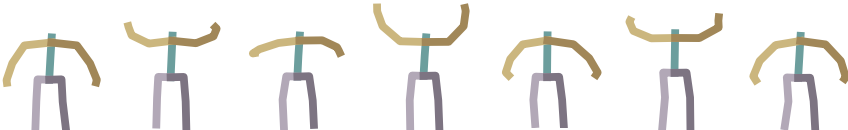
EHE
IRDS

KERAAL
SPHERE

KIMORE
UCDHE

KINECAL
UI-PRMD

- ▷ **Each exercise is treated as a separate dataset**



Category	# Datasets	# Sequences
Classification	39	14 721
Extrinsic regression	21	2 550
Total	60	17 291

The Rehab-Pile

- ▷ We aggregated 8 existing repositories :

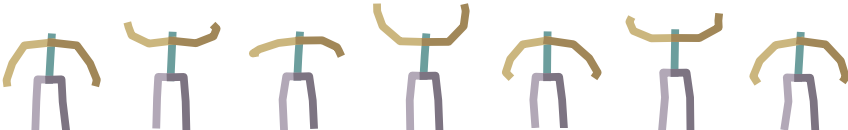
EHE
IRDS

KERAAL
SPHERE

KIMORE
UCDHE

KINECAL
UI-PRMD

- ▷ **Each exercise is treated as a separate dataset**



Category	# Datasets	# Sequences
Classification	39	14 721
Extrinsic regression	21	2 550
Total	60	17 291

Diverse Deep Learning Architectures

- ▷ Various existing architectures proposed for time series or human motion



Convolution
H-Inception
LITEMV



Recurrent
MotionGru
ConvLSTM



Graph
STGCN



Transformer
Vanilla Transformer
ConvTran

- ▷ **Ensembles of 5 models randomly initialized**

Experimental Protocol

Cross-subject evaluation

- ▷ Test sets contain **only unhealthy** subjects whenever possible
- ▷ 5-fold if # subjects ≥ 10
- ▷ Leave-One-Subject-Out if # subjects < 10

Metrics

▷ Classification :

- ▷ Accuracy
- ▷ Balanced Accuracy

▷ Extrinsic regression :

- ▷ RMSE
- ▷ MAE

Efficiency :

- ▷ Training time
- ▷ Inference time
- ▷ FLOPs
- ▷ Number of parameters

Experimental Protocol

Cross-subject evaluation

- ▷ Test sets contain **only unhealthy** subjects whenever possible
- ▷ 5-fold if # subjects ≥ 10
- ▷ Leave-One-Subject-Out if # subjects < 10

Metrics

▷ Classification :

- ▷ Accuracy
- ▷ Balanced Accuracy

▷ Extrinsic regression :

- ▷ RMSE
- ▷ MAE

Efficiency :

- ▷ Training time
- ▷ Inference time
- ▷ FLOPs
- ▷ Number of parameters

Classification Results

Model	Accuracy (%)	Balanced Accuracy (%)
STGCN	64.55 ± 14.91	57.65 ± 10.68
ConvLSTM	67.08 ± 16.37	62.59 ± 13.88
ConvTran	70.61 ± 12.90	65.72 ± 09.69
H-Inception	70.78 ± 12.75	65.67 ± 10.72
MotionGRU	72.82 ± 12.22	<u>67.58 ± 11.88</u>
VanTran	<u>74.30 ± 13.34</u>	67.47 ± 13.79
LITEMV	75.49 ± 11.28	70.34 ± 11.18

Classification Results

Model	Accuracy (%)	Balanced Accuracy (%)
STGCN	64.55 ± 14.91	57.65 ± 10.68
ConvLSTM	67.08 ± 16.37	62.59 ± 13.88
ConvTran	70.61 ± 12.90	65.72 ± 09.69
H-Inception	70.78 ± 12.75	65.67 ± 10.72
MotionGRU	72.82 ± 12.22	<u>67.58 ± 11.88</u>
VanTran	<u>74.30 ± 13.34</u>	67.47 ± 13.79
LITEMV	75.49 ± 11.28	70.34 ± 11.18

▷ MotionGru, VanTran and LITEMV are more accurate

Extrinsic Regression Results

Model	RMSE	MAE
STGCN	8.1084 ± 12.1624	6.7389 ± 10.2665
ConvLSTM	7.9354 ± 11.9593	6.5461 ± 10.0711
ConvTran	<u>7.3491 ± 10.8790</u>	<u>6.0729 ± 09.0227</u>
H-Inception	7.9342 ± 11.7514	6.3744 ± 09.3453
MotionGRU	7.8016 ± 11.6802	6.2966 ± 09.5963
VanTran	8.1758 ± 12.5573	6.8688 ± 10.5411
LITEMV	6.4286 ± 09.6731	5.3497 ± 08.1396

Extrinsic Regression Results

Model	RMSE	MAE
STGCN	8.1084 ± 12.1624	6.7389 ± 10.2665
ConvLSTM	7.9354 ± 11.9593	6.5461 ± 10.0711
ConvTran	<u>7.3491 ± 10.8790</u>	<u>6.0729 ± 09.0227</u>
H-Inception	7.9342 ± 11.7514	6.3744 ± 09.3453
MotionGRU	7.8016 ± 11.6802	6.2966 ± 09.5963
VanTran	8.1758 ± 12.5573	6.8688 ± 10.5411
LITEMV	6.4286 ± 09.6731	5.3497 ± 08.1396

- ▷ ConvTran and LITEMV perform better
 - ▷ Local temporal features

Efficiency Results

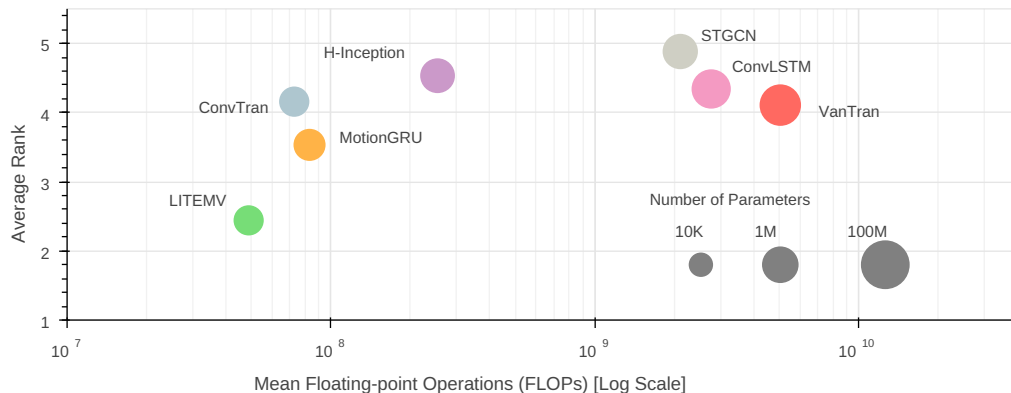
	Training time (min.)	Inference time (sec.)	# params (Million)	FLOPs (Billion)
STGCN	67.8 ± 99.9	1.2 ± 0.7	0.60 ± 0.90	2.1 ± 2.3
ConvLSTM	02.6 ± 01.9	<u>0.2 ± 0.1</u>	2.70 ± 1.40	2.8 ± 2.9
ConvTran	00.9 ± 00.5	0.5 ± 0.1	<u>0.09 ± 0.04</u>	<u>0.1 ± 0.1</u>
H-Inception	15.5 ± 05.1	0.9 ± 0.2	0.50 ± 0.01	0.3 ± 0.3
MotionGRU	<u>01.2 ± 01.1</u>	0.1 ± 0.0	0.20 ± 0.02	0.1 ± 0.1
VanTran	03.2 ± 03.0	1.5 ± 0.2	6.30 ± 0.01	5.1 ± 6.9
LITEMV	02.0 ± 01.7	0.6 ± 0.2	0.08 ± 0.05	0.1 ± 0.0

Efficiency Results

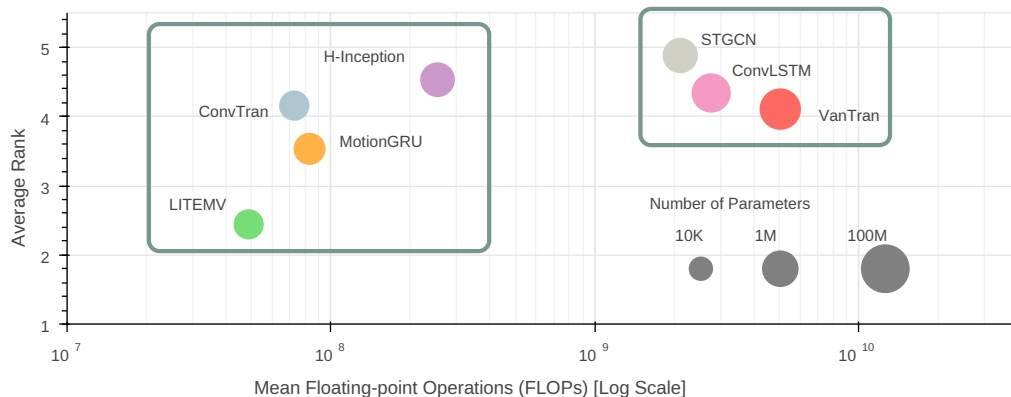
	Training time (min.)	Inference time (sec.)	# params (Million)	FLOPs (Billion)
STGCN	67.8 ± 99.9	1.2 ± 0.7	0.60 ± 0.90	2.1 ± 2.3
ConvLSTM	02.6 ± 01.9	<u>0.2 ± 0.1</u>	2.70 ± 1.40	2.8 ± 2.9
ConvTran	00.9 ± 00.5	0.5 ± 0.1	<u>0.09 ± 0.04</u>	<u>0.1 ± 0.1</u>
H-Inception	15.5 ± 05.1	0.9 ± 0.2	0.50 ± 0.01	0.3 ± 0.3
MotionGRU	<u>01.2 ± 01.1</u>	0.1 ± 0.0	0.20 ± 0.02	0.1 ± 0.1
VanTran	03.2 ± 03.0	1.5 ± 0.2	6.30 ± 0.01	5.1 ± 6.9
LITEMV	02.0 ± 01.7	0.6 ± 0.2	0.08 ± 0.05	0.1 ± 0.0

- ▷ ConvTran and MotionGru are faster
- ▷ LITEMV is smaller

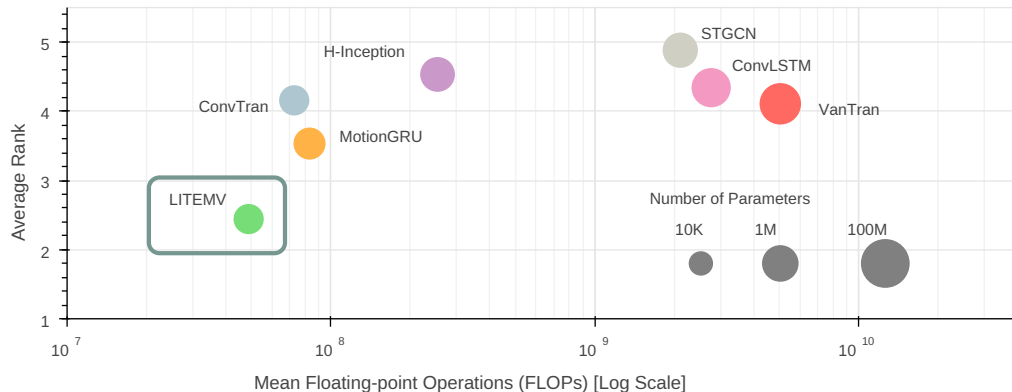
Performance vs Efficiency Trade-off



Performance vs Efficiency Trade-off



Performance vs Efficiency Trade-off



- ▷ Best overall trade-off :
 - ▷ **LITEMV**

Conclusion and Future Work

Conclusion

- ▷ A new unified framework for skeleton-based rehabilitation assessment
 - ▷ 7 deep learning architectures are compared
 - ▷ Rehab-Pile archive from 8 existing repositories
 - ▷ 39 datasets for classification
 - ▷ 21 datasets for extrinsic regression

Future work

- ▷ Our framework can serve as a reference for evaluating further approaches
- ▷ Facilitate the training of a dedicated foundation model

Conclusion and Future Work

Conclusion

- ▷ A new unified framework for skeleton-based rehabilitation assessment
 - ▷ 7 deep learning architectures are compared
 - ▷ Rehab-Pile archive from 8 existing repositories
 - ▷ 39 datasets for classification
 - ▷ 21 datasets for extrinsic regression

Future work

- ▷ Our framework can serve as a reference for evaluating further approaches
- ▷ Facilitate the training of a dedicated foundation model

Reproducibility and extension of our work

Don't hesitate to contribute :)



New model ?

Add a new model
and compare it under
the same protocol



New dataset ?

Add a new rehabilitation
dataset and evaluate
models on it



New metric ?

Add new evaluation criteria
and compare existing
models

```
$ ./run_regression_experiments.sh  
$ ./run_classification_experiments.sh
```

Code



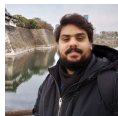
Data



<https://github.com/MSD-IRIMAS/DeepRehabPile>

<https://msd-irimas.github.io/pages/DeepRehabPile/>

The team



Ali
Ismail-Fawaz



Maxime
Devanne



Stefano
Berretti



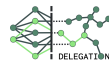
Jonathan
Weber



Germain
Forestier

Funding

ANR DELEGATION project (grant ANR-21-CE23-0014)



Included repositories in Rehab-Pile

EHE

X. Bruce, Y. Liu, K. C. Chan, Q. Yang, and X. Wang. Skeleton-based human action evaluation using graph convolutional network for monitoring alzheimer's progression. *Pattern Recognition*, 119 :108095, 2021.

KERAAL

S. M. Nguyen, M. Devanne, O. Remy-Neris, M. Lempereur, and A. Thepaut. A medical low-back pain physical rehabilitation database for human body movement analysis. In *International Joint Conference on Neural Networks*, 2024.

KIMORE

M. Capecci, M. G. Ceravolo, F. Ferracuti, S. Iarlori, A. Monteriu, L. Romeo, and F. Verdini. The kimore dataset : Kinematic assessment of movement and clinical scores for remote monitoring of physical rehabilitation. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 27(7) :1436–1448, 2019.

KINECAL

S. Maudsley-Barton and M. H. Yap. Kinecal : a dataset for falls-risk assessment and balance impairment analysis. *Scientific data*, 10(1) :633, 2023.

Included repositories in Rehab-Pile

IRDS

A. Miron, N. Sadawi, W. Ismail, H. Hussain, and C. Grosan. Intellirehabds (irds)—a dataset of physical rehabilitation movements. *Data*, 6(5) :46, 2021.

SPHERE

A. T. Paiement, L. Tao, S. L. Hannuna, M. Camplani, D. Damen, and M. Mirmehdi. Online quality assessment of human movement from skeleton data. In *British Machine Vision Conference*, 2014.

UCDHE

A. Singh, A. Bevilacqua, T. B. Aderinola, T. L. Nguyen, D. Whelan, M. O'Reilly, B. Caulfield, and G. Ifrim. An examination of wearable sensors and video data capture for human exercise classification. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 312–329. Springer, 2023.

UI-PRMD

A. Vakanski, H.-p. Jun, D. Paul, and R. Baker. A data set of human body movements for physical rehabilitation exercises. *Data*, 3(1) :2, 2018.

Included DL models

H-Inception

A. Ismail-Fawaz, M. Devanne, J. Weber, and G. Forestier. Deep learning for time series classification using new hand-crafted convolution filters. In IEEE Int. Conf. on Big Data, pages 972–981. IEEE, 2022

LITEMV

A. Ismail-Fawaz, M. Devanne, S. Berretti, J. Weber, and G. Forestier. Look into the lite in deep learning for time series classification. Int. Journal of Data Science and Analytics, pages 1–21, 2025.

ConvLSTM

F. J. Ordonez and D. Roggen. Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition. Sensors, 16(1) :115, 2016.

MotionGRU

H. Wu, Z. Yao, J. Wang, and M. Long. Motionrnn : A flexible model for video prediction with spacetime-varying motions. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pages 15435–15444, 2021.

Included DL models

STGCN

S. Deb, M. F. Islam, S. Rahman, and S. Rahman. Graph convolutional networks for assessment of physical rehabilitation exercises. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30 :410–419, 2022.

Vanilla Transformer

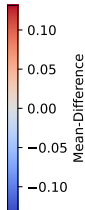
M. Petrovich, M. J. Black, and G. Varol. Action-conditioned 3d human motion synthesis with transformer vae. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10985–10995, 2021.

ConvTran

N. M. Foumani, C. W. Tan, G. I. Webb, and M. Salehi. Improving position encoding of transformers for multivariate time series classification. *Data mining and knowledge discovery*, 38(1) :22–48, 2024.

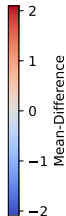
MCM - Classification (Accuracy)

	LITEMV 0.7549	VanTran 0.7430	MotionGRU 0.7282	H-Inception 0.7078	ConvTran 0.7061	ConvLSTM 0.6708	STGCN 0.6455
Mean-Accuracy							
LITEMV 0.7549	Mean-Difference $r > c / r = c / r < c$ Wilcoxon p-value	0.0119 22 / 1 / 16 0.4304	0.0267 26 / 1 / 12 0.0300	0.0470 29 / 1 / 9 0.0001	0.0487 33 / 0 / 6 0.0001	0.0840 28 / 0 / 11 0.0003	0.1094 30 / 0 / 9 $\leq 1e-04$
VanTran 0.7430	-0.0119 16 / 1 / 22 0.4304	-	0.0148 21 / 1 / 17 0.2613	0.0351 22 / 0 / 17 0.1711	0.0368 23 / 2 / 14 0.1230	0.0721 23 / 1 / 15 0.0060	0.0975 27 / 0 / 12 0.0002
MotionGRU 0.7282	-0.0267 12 / 1 / 26 0.0300	-0.0148 17 / 1 / 21 0.2613	-	0.0204 20 / 1 / 18 0.3907	0.0220 19 / 0 / 20 0.4063	0.0574 25 / 2 / 12 0.0030	0.0827 27 / 1 / 11 0.0002
H-Inception 0.7078	-0.0470 9 / 1 / 29 0.0001	-0.0351 17 / 0 / 22 0.1711	-0.0204 18 / 1 / 20 0.3907	-	0.0017 17 / 1 / 21 0.9722	0.0370 21 / 1 / 17 0.1336	0.0624 25 / 0 / 14 0.0065
ConvTran 0.7061	-0.0487 6 / 0 / 33 0.0001	-0.0368 14 / 2 / 23 0.1230	-0.0220 20 / 0 / 19 0.4063	-0.0017 21 / 1 / 17 0.9722	-	0.0353 22 / 1 / 16 0.1693	0.0607 26 / 1 / 12 0.0122
ConvLSTM 0.6708	-0.0840 11 / 0 / 28 0.0003	-0.0721 15 / 1 / 23 0.0060	-0.0574 12 / 2 / 25 0.0030	-0.0370 17 / 1 / 21 0.1336	-0.0353 16 / 1 / 22 0.1693	-	0.0254 20 / 2 / 17 0.2794
STGCN 0.6455	-0.1094 9 / 0 / 30 $\leq 1e-04$	-0.0975 12 / 0 / 27 0.0002	-0.0827 11 / 1 / 27 0.0002	-0.0624 14 / 0 / 25 0.0065	-0.0607 12 / 1 / 26 0.0122	-0.0254 17 / 2 / 20 0.2794	If in bold, then p-value < 0.05

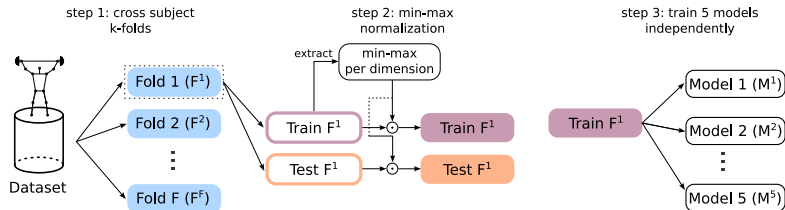


MCM - Extrinsic regression (RMSE)

	LITEMV 6.4286	ConvTran 7.3491	MotionGRU 7.8016	H-Inception 7.9342	ConvLSTM 7.9354	STGCN 8.1084	VanTran 8.1758
Mean-R-MSE							
LITEMV 6.4286	Mean-Difference r<c / r=c / r>c Wilcoxon p-value	-0.9205 19 / 0 / 2 ≤ 1e-04	-1.3730 17 / 0 / 4 0.0014	-1.5056 19 / 0 / 2 0.0001	-1.5068 18 / 0 / 3 ≤ 1e-04	-1.6798 16 / 0 / 5 0.0012	-1.7471 17 / 0 / 4 0.0014
ConvTran 7.3491	0.9205 2 / 0 / 19 ≤ 1e-04	-	-0.4526 9 / 0 / 12 1.0000	-0.5852 13 / 0 / 8 0.2290	-0.5864 11 / 0 / 10 0.6578	-0.7593 12 / 0 / 9 0.4948	-0.8267 14 / 0 / 7 0.2572
MotionGRU 7.8016	1.3730 4 / 0 / 17 0.0014	0.4526 12 / 0 / 9 1.0000	-	-0.1326 15 / 0 / 6 0.0547	-0.1338 13 / 0 / 8 0.1032	-0.3068 17 / 0 / 4 0.0014	-0.3741 14 / 0 / 7 0.1790
H-Inception 7.9342	1.5056 2 / 0 / 19 0.0001	0.5852 8 / 0 / 13 0.2290	0.1326 6 / 0 / 15 0.0547	-	-0.0012 7 / 0 / 14 0.2877	-0.1742 10 / 0 / 11 1.0000	-0.2415 11 / 0 / 10 0.8917
ConvLSTM 7.9354	1.5068 3 / 0 / 18 ≤ 1e-04	0.5864 10 / 0 / 11 0.6578	0.1338 8 / 0 / 13 0.1032	0.0012 14 / 0 / 7 0.2877	-	-0.1730 13 / 0 / 8 0.1373	-0.2403 14 / 0 / 7 0.1571
STGCN 8.1084	1.6798 5 / 0 / 16 0.0012	0.7593 9 / 0 / 12 0.4948	0.3068 4 / 0 / 17 0.0014	0.1742 11 / 0 / 10 1.0000	0.1730 8 / 0 / 13 0.1373	-	-0.0674 14 / 0 / 7 0.7593
VanTran 8.1758	1.7471 4 / 0 / 17 0.0014	0.8267 7 / 0 / 14 0.2572	0.3741 7 / 0 / 14 0.1790	0.2415 10 / 0 / 11 0.8917	0.2403 7 / 0 / 14 0.1571	0.0674 7 / 0 / 14 0.7593	If in bold, then p-value < 0.05



General pipeline



$\odot$: min-max normalization

